

# AA411

## Forces Test Procedure

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**Team Name:** ROFASO

**Project Name:** Damage Assessment Protocol (D.A.P.)

**Sub-System Name:** Plant: Forces

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## 1 Objective of Test Procedure

The goal of this procedure is to verify that the Seaglider forces code equations correctly calculate the lift and drag forces upon the Seaglider from the computational fluid dynamics (CFD) coefficient outputs, and are within a 3.30% error bound of the hydrodynamic model (HDM). The buoyancy force is directly taken from the HDM and does not need to be tested as it is now an input into the whole Digital Twin system.

The outputs of this test are lift (L), drag (D), and the percent difference between the calculated lift and drag from the CFD model and the calculated lift and drag from the HDM.

The following are the requirements that will be tested using this procedure.

- **FMR-1:** The Forces Modeler shall compute the lift force  $\leq 3.30\%$  error of the lift force from the existing nominal planar flight hydrodynamic model.
- **FMR-2:** The Forces Modeler shall compute the drag force  $\leq 3.30\%$  error of the drag force from the existing nominal planar flight hydrodynamic model.
- **FMR-3:** The Forces Modeler shall compute buoyant force  $\leq 3.30\%$  error of buoyancy estimator on board of the Seaglider

## 2 Measurements

| Variable | Description   | Units | Measurement Method     |
|----------|---------------|-------|------------------------|
| $L$      | Lift force    | N     | Forces function output |
| $D$      | Drag Force    | N     | Forces function output |
| $B$      | Buoyant Force | N     | Forces function output |

## 3 Equipment Required

| Equipment/Tools | Description                                            | Qty | Check |
|-----------------|--------------------------------------------------------|-----|-------|
| Laptop          | At least 8 GB RAM and 256 GB SSD is required           | 1   |       |
| Google Sheets   | Used to calculate and graph lift/drag force versus AoA | 1   |       |

## 4 Procedure: Lift and Drag Forces

### 1. Setup

1. Access the Google sheet file *CFD Post Processing* from the ROFASO Google Drive.
  - (a) AA - Resilient Ocean Flight Autonomy for Seaglider Operation  $\rightarrow$  StarCCM+  $\rightarrow$  CFD Post Processing.
2. Find the  $C_L$  and  $C_D$  coefficients for planar flight at all AoA, e.g.  $\beta = 0$ .
3. Access the Google Sheet file *Forces Test - Lift and Drag*.
  - (a) AA - Resilient Ocean Flight Autonomy for Seaglider Operation  $\rightarrow$  Meeting Agenda and Slides  $\rightarrow$  Forces  $\rightarrow$  Forces Test - Lift and Drag

**Pass/Fail Criteria:** Access to the test documents

## 2. Calculating Force of Lift and Drag from CFD

1. Open *Forces Test - Lift and Drag*
2. Input the AoA ( $\alpha$ ),  $C_L$ , and  $C_D$  (for  $\beta = 0$ ) coefficient information into the first three columns of the Google Sheet
3. Use the equations for lift force (L) and drag force (D) from the dynamicsV3 code and calculate the L and D for each corresponding AoA

$$L = qSC_L$$

$$D = qSC_D$$

4. Graph both the lift and the drag force against AoA

## 3. Calculating Force of Lift and Drag from Hydrodynamic Model Equations (Bennett et al.)

1. Keep the *Forces Test - Lift and Drag* Google Sheet open
2. Using the same AoA sweep used in **Section 2**, calculate L and D using the hydrodynamic model equations

$$L = qS\alpha a$$

$$D = qS(bq^{1/4} + c\alpha^2)$$

## 4. Calculating Percent Difference Between CFD and HDM

1. In the same Google Sheet, calculate the percentage difference between the two forces calculated from CFD and calculated from HDM for all AoA

$$\%diff = \frac{|CFD - HDM|}{|HDM|} \cdot 100$$

**Pass/Fail Criteria:** The difference between the lift, drag of CFD and HDM are  $\leq 3.30\%$

| Criteria | CHECK |
|----------|-------|
| PASS     |       |
| FAIL     |       |

## 5 Shut Down

1. Save *Forces Test - Lift and Drag*
  - (a) Saved? Yes/No \_\_\_\_\_
2. Shut down laptop
  - (a) Power Off? Yes/No \_\_\_\_\_